

Tungsten Carbide End Mills UNIMAX Series

Vol. 5

HMGCOAT 2 Flutes Ball End Mills

Add 2

HGB

Total 20 Models

For Hard Materials

HMGCOAT 2 Flutes Long Neck Ball End Mills

Add 7

HGLB

Total 101 Models



UNION TOOL CO.

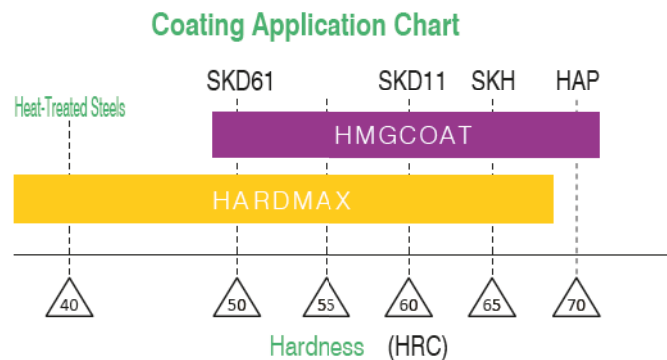
HGB / HGLB Features of HGB / HGLB

1 Adopted new coating "HMGCOAT"

Newly developed "HMGCOAT" (HARDMAX GREAT COAT)

High hardness coating offers higher wear resistance compared to HARDMAX.

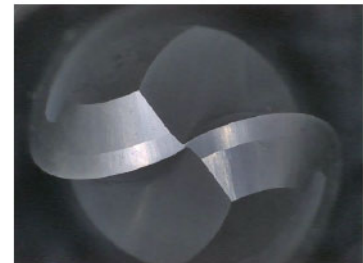
Long tool life for milling hardened materials over 60 HRC.



2 New Carbide Materials with excellent wear resistance

3 New design with improved breakage resistance

The rake and relief angles optimized for milling hard materials improve breakage resistance.



4 High Precision Diameter Tolerance / Radius Accuracy / Shank Diameter Tolerance

HSB / HSLB Tolerance

Radius of Ball Nose	Diameter Tolerance	Ball Radius Accuracy	Shank Diameter Tolerance
R0.03 ~ R0.075	0/-0.01	± 0.002	0/-0.005 (h5)
R0.1 ~ R3	0/-0.015	± 0.005	

* Only HSB1001-0020-6 is 1 flute.

Ball Radius Accuracy: ±0.005, Diameter Tolerance: 0/-0.015

HGB / HGLB Tolerance

Radius of Ball Nose	Diameter Tolerance	Ball Radius Accuracy	Shank Diameter Tolerance	
R0.05 ~ R0.075	0/-0.008	± 0.002	0/-0.004 (h4)	
R0.1 ~ R1.25		± 0.003		
R1.5 ~ R2	0/-0.01	± 0.005		
R2.5 ~ R3				

Shank diameter tolerance h4!

HMGCOAT 2 Flute Ball End Mills for Hard Materials



Size **R0.05~R3**

HGB

Super
MG

HMG
COAT

30°

R
±0.002
R0.05~R0.075

R
±0.003
R0.1~R2

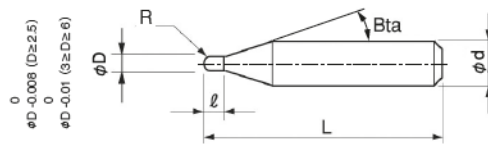
R
±0.005
R2.5~R3

Shank Dia
0/-0.004

Additional 2 Models

Material Applications (☆ Highly Recommended ◎ Recommended ○ Suggested)

Work Material															
CARBON STEELS S45C S55C	ALLOY STEELS SK / SCM SUS	PREHARDENED STEELS NAK HPM	HARDENED STEELS			CAST IRON	ALUMINUM ALLOYS	GRAPHITE	COPPER	PLASTICS	GLASS FILLED PLASTICS	TITANIUM ALLOYS	HEAT RESISTANT ALLOYS	CEMENTED CARBIDE	HARD BRITTLE (NON-METALLIC) MATERIALS
			~ 55HRC	~ 60HRC	~ 70HRC										
		○	◎	◎	◎										



The shank taper angle shown is not an exact value and to avoid contact with the workpiece, we recommend the user controls the precise value of this angle. Shank taper angle should not make contact with the work piece.

Total 20 models

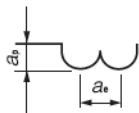
Unit (mm)

Model Number	Radius of Ball Nose R	Length of Cut ℓ	Shank Taper Angle Bta	Overall Length L	Shank Diameter Ø d	Price ¥
* HGB 2001-0010	R0.05	0.1	16°	50	4	12,960
* HGB 20015-0015	R0.075	0.15	16°	50	4	12,600
HGB 2002-0030	R0.1	0.3	16°	50	4	9,120
HGB 2003-0030	R0.15	0.3	16°	50	4	7,440
HGB 2003-0045		0.45		50	4	7,440
HGB 2004-0040	R0.2	0.4	16°	50	4	5,040
HGB 2004-0060		0.6		50	4	5,040
HGB 2005-0050	R0.25	0.5	16°	50	4	4,680
HGB 2005-0075		0.75		50	4	4,680
HGB 2006-0060	R0.3	0.6	16°	50	4	4,560
HGB 2006-0090		0.9		50	4	4,560
HGB 2008-0120	R0.4	1.2	16°	50	4	4,560
HGB 2010-0150	R0.5	1.5	16°	50	4	4,150
HGB 2015-0225	R0.75	2.25	16°	50	4	5,040
HGB 2020-0300	R1	3	16°	50	4	3,720
HGB 2025-0375	R1.25	3.75	16°	50	4	6,370
HGB 2030-0450	R1.5	4.5	16°	50	6	4,560
HGB 2040-0600	R2	6	16°	50	6	5,160
HGB 2050-0750	R2.5	7.5	16°	50	6	6,240
HGB 2060-0900	R3	9	16°	50	6	6,480

* Additional model

HGB Milling Conditions

WORK MATERIAL			PREHARDENED STEELS / HARDENED STEELS NAK / STAVAX (~55HRC)				HARDENED STEELS SKD11 (55~62HRC)				HARDENED STEELS HAP10 (62~68HRC)				HARDENED STEELS HAP72 (68~70HRC)			
Model Number	Radius of Ball Nose (mm)	Length of Cut (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)
2001-0010	R0.05	0.1	48,000	200	0.005	0.01	48,000	200	0.005	0.01	48,000	150	0.003	0.006	40,000	120	0.002	0.004
20015-0015	R0.075	0.15	48,000	230	0.007	0.014	48,000	230	0.007	0.014	48,000	170	0.005	0.01	40,000	135	0.003	0.006
2002-0030	R0.1	0.3	44,000	250	0.01	0.03	42,000	250	0.01	0.03	40,000	200	0.008	0.024	36,000	150	0.006	0.018
2003-0030	R0.15	0.3	44,000	400	0.01	0.03	42,000	350	0.01	0.03	40,000	300	0.01	0.03	36,000	250	0.008	0.024
2003-0045		0.45	44,000	400	0.01	0.03	42,000	350	0.01	0.03	40,000	300	0.01	0.03	36,000	250	0.008	0.024
2004-0040	R0.2	0.4	44,000	600	0.015	0.045	42,000	550	0.015	0.045	40,000	500	0.013	0.036	36,000	350	0.01	0.027
2004-0060		0.6	44,000	600	0.015	0.045	42,000	550	0.015	0.045	40,000	500	0.013	0.036	36,000	350	0.01	0.027
2005-0050	R0.25	0.5	44,000	900	0.02	0.065	40,000	800	0.015	0.05	36,000	600	0.015	0.05	30,000	400	0.015	0.03
2005-0075		0.75	44,000	900	0.02	0.065	40,000	800	0.015	0.05	36,000	600	0.015	0.05	30,000	400	0.015	0.03
2006-0060	R0.3	0.6	40,000	1,400	0.045	0.15	36,000	1,200	0.025	0.13	32,000	1,000	0.02	0.1	25,000	600	0.02	0.1
2006-0090		0.9	40,000	1,400	0.045	0.15	36,000	1,200	0.025	0.13	32,000	1,000	0.02	0.1	25,000	600	0.02	0.1
2008-0120	R0.4	1.2	35,000	1,600	0.06	0.21	30,000	1,600	0.04	0.17	26,000	1,350	0.04	0.15	20,000	700	0.02	0.12
2010-0150	R0.5	1.5	30,000	1,750	0.2	0.4	24,000	2,000	0.1	0.3	21,000	1,750	0.05	0.2	16,000	875	0.05	0.2
2015-0225	R0.75	2.25	30,000	2,450	0.25	0.55	17,000	2,000	0.12	0.4	15,000	1,750	0.06	0.29	11,250	875	0.06	0.29
2020-0300	R1	3	28,000	2,900	0.3	0.7	14,000	2,100	0.15	0.5	14,700	2,160	0.1	0.35	11,040	1,080	0.08	0.35
2025-0375	R1.25	3.75	24,500	2,950	0.35	0.85	12,250	2,150	0.17	0.6	12,840	2,220	0.12	0.45	9,660	1,110	0.1	0.45
2030-0450	R1.5	4.5	21,000	3,000	0.4	1	10,500	2,200	0.2	0.7	11,040	2,280	0.15	0.55	8,280	1,140	0.12	0.55
2040-0600	R2	6	18,000	3,200	0.5	1.3	9,000	2,300	0.25	0.95	9,480	2,400	0.18	0.75	7,080	1,200	0.15	0.75
2050-0750	R2.5	7.5	15,600	3,500	0.5	1.5	7,800	2,500	0.25	1.05	8,160	2,520	0.2	0.85	6,120	1,260	0.15	0.85
2060-0900	R3	9	13,000	3,500	0.6	1.8	6,500	2,500	0.3	1.3	6,840	2,640	0.25	1	5,000	1,500	0.2	1



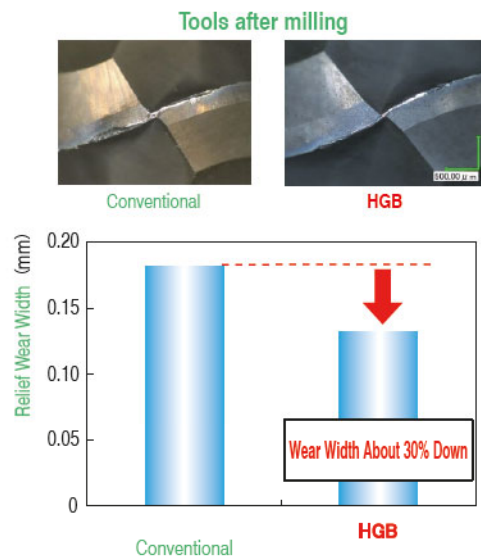
Note:

- Decrease the feed rate more than 50% from the milling parameters when slot milling.
- Decrease both spindle speed and feed rate proportionally when the milling parameters exceed the machine's maximum spindle speed, or when chattering and red-hot occur.
- Every coolant offers stable milling.

Wear Comparison

HGB R3 × Length of Cut 9

HAP10 (64HRC)

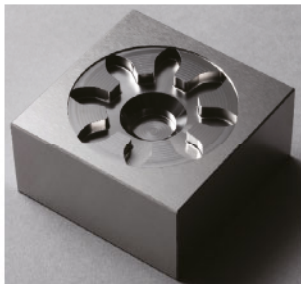


Tool	HGB 2060-0900
Spindle Speed	6,840 min ⁻¹
Feed Rate	2,640 mm/min
Axial Depth a_p	0.25 mm
Radial Depth a_e	1 mm
Coolant	Air Blow (Through-Spindle)
Milling Shape	Square Pocket (55 x 17.5 x Depth 6.9 mm) × 5 Pockets
Cycle Time	75 min

Milling Example of Spur Gear Die

HGB / HGLB R0.5 • R1 • R1.5

HAP72 (68HRC)



Work Size : 60 x 60 x 30 mm
Milling Size : Ø 50.4 x Depth 11 mm
Coolant : Air Blow (Through-Spindle)



HGB / HGLB
Spur Gear Die
Milling Video

No.	Process	Tool	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	Axial Depth (mm)	Radial Depth (mm)	Allowance (mm)	Cycle Time (h:m)
1	Roughing	HGB 2030-0450 (R1.5 x Length of Cut 4.5)	8,280	1,140	0.12	0.55	0.02	1:41
2	Roughing	HGLB 2020-060 (R1 x Effective Length 6)	12,250	1,800	0.06	0.3	0.02	0:06
3	Semi-finishing		12,250	1,800	0.06	0.05	0.01	0:53
4	Finishing		12,250	900	0.00015 Cusp Height	-	0	0:09
5	Finishing	HMERS 4030-01-0750 (Ø 3 x CR0.1 x Length of Cut 7.5)	8,600	465 / 1,500	0.5	1	0	0:02
6	Finishing		8,600	465 / 1,500	0.0002 Cusp Height	1	0	0:01
7	Finishing	HGLB 2010-060 (R0.5 x Effective Length 6)	20,000	800	0.00015 Cusp Height	-	0	0:55
8	Finishing		20,000	800	0.01	0.04	0	0:51

Total 4:38

HMGCOAT 2 Flute Long Neck Ball End Mills for Hard Materials



Size **R0.05~R3**

HGLB

Super
MG

HMG
COAT

30°

R
±0.002
R0.05~R0.075

R
±0.003
R0.1~R2

R
±0.005
R2.5~R3

Shank Dia
0/-0.004

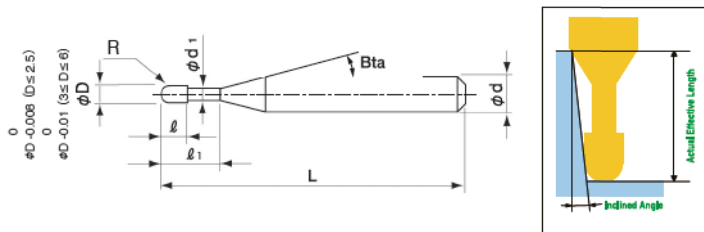
Back Taper
Geometry

Back taper geometry does not apply to
R0.4 or below, $\epsilon_1 / D \leq 10$.

Additional 7 Models

Material Applications (☆ Highly Recommended ◎ Recommended ○ Suggested)

Work Material															
CARBON STEELS S45C S55C	ALLOY STEELS SK / SCM SUS	PREHARDENED STEELS NAK HPM	HARDENED STEELS			CAST IRON	ALUMINUM ALLOYS	GRAPHITE	COPPER	PLASTICS	GLASS FILLED PLASTICS	TITANIUM ALLOYS	HEAT RESISTANT ALLOYS	CEMENTED CARBIDE	HARD BRITTLE (NON-METALLIC) MATERIALS
			~ 55HRC	~ 60HRC	~ 70HRC										
		○	◎	◎	◎										



The shank taper angle shown is not an exact value and to avoid contact with the workpiece, we recommend the user controls the precise value of this angle. Shank taper angle should not make contact with the work piece.

Total 101 models

Unit (mm)

Model Number	Radius of Ball Nose R	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter ϕd_1	Shank Taper Angle Bta	Overall Length L	Shank Diameter ϕd	Price ¥	Effective Length by Inclined Angles				
									30°	1°	1°30'	2°	3°
* HGLB 2001-002	R0.05	0.2	0.08	0.095	16°	45	4	12,360	0.22	0.24	0.26	0.28	0.31
* HGLB 2001-003		0.3				45	4	12,360	0.33	0.36	0.38	0.40	0.44
* HGLB 2001-005		0.5				45	4	13,440	0.55	0.58	0.61	0.64	0.68
* HGLB 20015-003	R0.075	0.3	0.12	0.14	16°	45	4	14,400	0.35	0.37	0.39	0.41	0.44
* HGLB 20015-005		0.5				45	4	15,240	0.56	0.59	0.62	0.64	0.69
* HGLB 20015-0075		0.75				45	4	16,020	0.83	0.86	0.90	0.93	1.00
* HGLB 20015-010		1				45	4	16,020	1.09	1.13	1.17	1.21	1.30
HGLB 2002-003	R0.1	0.3	0.16	0.19	16°	45	4	8,640	0.42	0.44	0.46	0.48	0.52
HGLB 2002-005		0.5				45	4	8,640	0.63	0.66	0.68	0.71	0.76
HGLB 2002-010		1				45	4	8,640	1.15	1.20	1.24	1.28	1.37
HGLB 2002-015		1.5				45	4	9,360	1.66	1.72	1.78	1.84	1.97
HGLB 2003-005	R0.15	0.5	0.24	0.29	16°	45	4	8,520	0.63	0.65	0.68	0.70	0.75
HGLB 2003-0075		0.75				45	4	8,520	0.89	0.92	0.96	0.99	1.05
HGLB 2003-010		1				45	4	8,520	1.15	1.19	1.23	1.27	1.36
HGLB 2003-015		1.5				45	4	9,120	1.66	1.72	1.77	1.83	1.96
HGLB 2003-020		2				45	4	9,120	2.18	2.25	2.32	2.40	2.57
HGLB 2003-030		3				45	4	9,360	3.21	3.32	3.42	3.54	3.80

* Additional model

HMGC0AT 2 Flute Long Neck Ball End Mills for Hard Materials

Model Number	Radius of Ball Nose R	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter $\varnothing d_1$	Shank Taper Angle $B\alpha$	Overall Length L	Shank Diameter $\varnothing d$	Price ¥	Effective Length by Inclined Angles				
									30°	1°	1°30'	2°	3°
HGLB 2004-005	R0.2	0.5	0.32	0.39	16°	45	4	5,880	0.63	0.65	0.67	0.70	0.74
HGLB 2004-0075		0.75				45	4	5,880	0.89	0.92	0.95	0.98	1.04
HGLB 2004-010		1				45	4	5,880	1.15	1.19	1.23	1.26	1.35
HGLB 2004-0125		1.25				45	4	6,000	1.40	1.45	1.49	1.54	1.64
HGLB 2004-015		1.5				45	4	6,000	1.66	1.71	1.77	1.82	1.95
HGLB 2004-020		2				45	4	6,120	2.18	2.25	2.32	2.39	2.56
HGLB 2004-025		2.5				45	4	6,360	2.70	2.78	2.87	2.96	3.17
HGLB 2004-030		3				45	4	6,720	3.21	3.31	3.42	3.53	3.79
HGLB 2004-040		4				45	4	7,320	4.24	4.38	4.52	4.67	5.01
HGLB 2005-010	R0.25	1	0.4	0.49	16°	45	4	5,880	1.15	1.19	1.22	1.26	1.34
HGLB 2005-015		1.5				45	4	5,880	1.65	1.71	1.76	1.82	1.94
HGLB 2005-020		2				45	4	5,880	2.18	2.24	2.31	2.39	2.55
HGLB 2005-025		2.5				45	4	5,880	2.69	2.78	2.86	2.96	3.16
HGLB 2005-030		3				45	4	5,880	3.21	3.31	3.41	3.53	3.77
HGLB 2005-040		4				45	4	5,880	4.24	4.37	4.51	4.66	5.00
HGLB 2005-050		5				45	4	6,000	5.27	5.44	5.61	5.80	6.22
HGLB 2005-080		6				45	4	6,120	6.30	6.50	6.71	6.94	7.45
HGLB 2006-010	R0.3	1	0.48	0.59	16°	45	4	5,040	1.14	1.18	1.22	1.25	1.33
HGLB 2006-015		1.5				45	4	4,560	1.65	1.71	1.76	1.81	1.93
HGLB 2006-020		2				45	4	4,560	2.17	2.24	2.31	2.38	2.54
HGLB 2006-025		2.5				45	4	4,680	2.69	2.77	2.86	2.95	3.15
HGLB 2006-030		3				45	4	4,680	3.21	3.31	3.41	3.52	3.76
HGLB 2006-040		4				45	4	4,800	4.24	4.37	4.51	4.66	4.99
HGLB 2006-050		5				45	4	4,800	5.27	5.44	5.61	5.80	6.21
HGLB 2006-080		6				45	4	4,800	6.30	6.50	6.71	6.93	7.43
HGLB 2006-080		8				45	4	6,360	8.37	8.63	8.91	9.21	9.88
HGLB 2008-020	R0.4	2	0.64	0.79	16°	45	4	4,560	2.17	2.23	2.30	2.37	2.52
HGLB 2008-040		4				45	4	4,800	4.24	4.36	4.50	4.64	4.97
HGLB 2008-060		6				45	4	4,800	6.30	6.49	6.70	6.92	7.41
HGLB 2008-080		8				45	4	4,800	8.36	8.62	8.90	9.20	9.86
HGLB 2010-020	R0.5	2	0.8	0.98	16°	45	4	3,840	2.18	2.24	2.30	2.36	2.51
HGLB 2010-025		2.5				45	4	3,840	2.70	2.77	2.85	2.93	3.12
HGLB 2010-030		3				45	4	3,840	3.21	3.30	3.40	3.50	3.73
HGLB 2010-040		4				45	4	4,320	4.24	4.37	4.50	4.64	4.96
HGLB 2010-050		5				45	4	4,320	5.28	5.43	5.60	5.78	6.18
HGLB 2010-080		6				45	4	4,680	6.31	6.50	6.70	6.92	7.40
HGLB 2010-080		8				45	4	4,680	8.37	8.63	8.90	9.20	9.85
HGLB 2010-100		10				45	4	4,680	10.43	10.76	11.10	11.47	12.30
HGLB 2010-120		12				45	4	4,680	12.50	12.89	13.30	13.75	14.75
HGLB 2010-140		14				45	4	5,400	14.56	15.02	15.51	16.03	17.19

HMGCOAT 2 Flute Long Neck Ball End Mills for Hard Materials

Model Number	Radius of Ball Nose R	Effective Length ℓ_1	Length of Cut ℓ	Neck Diameter $\varnothing d_1$	Shank Taper Angle $B\alpha$	Overall Length L	Shank Diameter $\varnothing d$	Price ¥	Effective Length by Inclined Angles				
									30°	1°	1°30'	2°	3°
HGLB 2015-030	R0.75	3	1.2	1.47	16°	45	4	4,440	3.10	3.18	3.26	3.35	3.55
HGLB 2015-040		4				45	4	4,440	4.13	4.24	4.36	4.49	4.77
HGLB 2015-060		6				45	4	4,440	6.19	6.37	6.56	6.76	7.22
HGLB 2015-080		8				45	4	4,680	8.25	8.50	8.76	9.04	9.67
HGLB 2015-100		10				45	4	5,040	10.32	10.63	10.96	11.32	12.11
HGLB 2015-120		12				45	4	5,400	12.38	12.76	13.16	13.60	14.56
HGLB 2015-140		14				45	4	5,400	14.44	14.89	15.36	15.87	17.01
HGLB 2020-030	R1	3	1.6	1.98	16°	45	4	3,840	3.07	3.14	3.21	3.29	3.47
HGLB 2020-040		4				45	4	3,840	4.10	4.20	4.31	4.43	4.70
HGLB 2020-040-6		4				50	6	6,120	4.10	4.20	4.31	4.43	4.70
HGLB 2020-060		6				45	4	4,320	6.16	6.33	6.51	6.71	7.14
HGLB 2020-060-6		6				50	6	6,600	6.16	6.33	6.51	6.71	7.14
HGLB 2020-080		8				45	4	4,680	8.23	8.46	8.72	8.99	9.59
HGLB 2020-080-6		8				50	6	7,080	8.23	8.46	8.72	8.99	9.59
HGLB 2020-100		10				45	4	4,680	10.29	10.59	10.92	11.26	12.04
HGLB 2020-100-6		10				50	6	7,080	10.29	10.59	10.92	11.26	12.04
HGLB 2020-120		12				45	4	4,680	12.35	12.72	13.12	13.54	14.48
HGLB 2020-120-6		12				50	6	7,080	12.35	12.72	13.12	13.54	14.48
HGLB 2020-140		14				45	4	4,680	14.41	14.85	15.32	15.82	16.93
HGLB 2020-160		16				45	4	4,680	16.48	16.98	17.52	18.10	19.38
HGLB 2020-200		20				60	4	4,680	20.60	21.24	21.92	22.65	No interference
HGLB 2030-060	R1.5	6	2.4	2.95	16°	60	6	4,680	6.20	6.35	6.52	6.69	7.09
HGLB 2030-080		8				60	6	4,680	8.26	8.48	8.72	8.97	9.54
HGLB 2030-100		10				60	6	5,400	10.32	10.61	10.92	11.25	11.99
HGLB 2030-120		12				60	6	5,640	12.38	12.74	13.12	13.53	14.43
HGLB 2030-140		14				60	6	6,240	14.45	14.87	15.32	15.80	16.88
HGLB 2030-160		16				60	6	6,240	16.51	17.00	17.52	18.08	19.33
HGLB 2030-200		20				70	6	6,000	20.64	21.26	21.92	22.64	24.22
HGLB 2030-250		25				70	6	6,000	25.79	26.58	27.43	28.33	30.34
HGLB 2030-300		30				70	6	6,840	30.95	31.91	32.93	34.02	No interference
HGLB 2040-080	R2	8	3.2	3.95	16°	70	6	4,800	8.24	8.45	8.67	8.90	9.43
HGLB 2040-100		10				70	6	4,800	10.31	10.58	10.87	11.18	11.88
HGLB 2040-120		12				70	6	6,240	12.37	12.71	13.07	13.46	14.32
HGLB 2040-140		14				70	6	6,240	14.43	14.84	15.27	15.74	16.77
HGLB 2040-160		16				70	6	6,240	16.49	16.97	17.47	18.01	19.22
HGLB 2040-200		20				70	6	6,240	20.62	21.23	21.87	22.57	No interference
HGLB 2040-250		25				70	6	6,240	25.78	26.55	27.38	28.26	No interference
HGLB 2040-300		30				70	6	6,240	30.93	31.87	32.88	No interference	No interference
HGLB 2040-350		35				80	6	7,200	36.09	37.20	38.38	No interference	No interference
HGLB 2060-100	R3	10	4.8	5.95	□	80	6	7,800	No interference	No interference	No interference	No interference	No interference
HGLB 2060-150		15				80	6	7,800	No interference	No interference	No interference	No interference	No interference
HGLB 2060-200		20				80	6	7,800	No interference	No interference	No interference	No interference	No interference
HGLB 2060-250		25				80	6	7,800	No interference	No interference	No interference	No interference	No interference
HGLB 2060-300		30				80	6	8,040	No interference	No interference	No interference	No interference	No interference

HGLB Milling Conditions

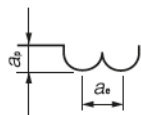
WORK MATERIAL			PREHARDENED STEELS / HARDENED STEELS NAK / STAVAX (~55HRC)				HARDENED STEELS SKD11 (55~62HRC)				HARDENED STEELS HAP10 (62~68HRC)				HARDENED STEELS HAP72 (68~70HRC)			
Model Number	Radius of Ball Nose (mm)	Effective Length (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)
2001-002	R0.05	0.2	48,000	200	0.005	0.01	48,000	200	0.005	0.01	48,000	150	0.003	0.006	40,000	120	0.002	0.004
2001-003		0.3	48,000	200	0.005	0.01	48,000	200	0.005	0.01	48,000	150	0.003	0.006	40,000	120	0.002	0.004
2001-005		0.5	48,000	200	0.005	0.01	48,000	200	0.005	0.01	48,000	150	0.003	0.006	40,000	120	0.002	0.004
20015-003	R0.075	0.3	48,000	230	0.007	0.014	48,000	230	0.007	0.014	48,000	170	0.005	0.01	40,000	135	0.003	0.006
20015-005		0.5	48,000	230	0.007	0.014	48,000	230	0.007	0.014	48,000	170	0.005	0.01	40,000	135	0.003	0.006
20015-0075		0.75	48,000	230	0.007	0.014	48,000	230	0.007	0.014	48,000	170	0.005	0.01	40,000	135	0.003	0.006
20015-010		1	38,400	160	0.005	0.01	38,400	160	0.005	0.01	38,400	120	0.003	0.007	32,000	90	0.002	0.004
2002-003	R0.1	0.3	44,000	250	0.01	0.03	42,000	250	0.01	0.03	40,000	200	0.008	0.024	36,000	150	0.006	0.018
2002-005		0.5	44,000	250	0.01	0.03	42,000	250	0.01	0.03	40,000	200	0.008	0.024	36,000	150	0.006	0.018
2002-010		1	44,000	250	0.01	0.03	42,000	250	0.01	0.03	40,000	200	0.008	0.024	36,000	150	0.006	0.018
2002-015		1.5	35,200	175	0.008	0.023	33,600	175	0.008	0.023	32,000	140	0.006	0.018	28,800	100	0.004	0.012
2003-005	R0.15	0.5	44,000	400	0.01	0.03	42,000	350	0.01	0.03	40,000	300	0.01	0.03	36,000	250	0.008	0.024
2003-0075		0.75	44,000	400	0.01	0.03	42,000	350	0.01	0.03	40,000	300	0.01	0.03	36,000	250	0.008	0.024
2003-010		1	44,000	400	0.01	0.03	42,000	350	0.01	0.03	40,000	300	0.01	0.03	36,000	250	0.008	0.024
2003-015		1.5	44,000	400	0.01	0.03	42,000	350	0.01	0.03	40,000	300	0.01	0.03	36,000	250	0.008	0.024
2003-020		2	35,200	280	0.008	0.023	33,600	245	0.008	0.023	32,000	210	0.008	0.023	28,800	175	0.006	0.018
2003-030		3	35,200	140	0.004	0.01	33,600	125	0.004	0.01	32,000	110	0.004	0.01	28,800	85	0.003	0.009
2004-005	R0.2	0.5	44,000	600	0.015	0.045	42,000	550	0.015	0.045	40,000	500	0.013	0.036	36,000	350	0.01	0.027
2004-0075		0.75	44,000	600	0.015	0.045	42,000	550	0.015	0.045	40,000	500	0.013	0.036	36,000	350	0.01	0.027
2004-010		1	44,000	600	0.015	0.045	42,000	550	0.015	0.045	40,000	500	0.013	0.036	36,000	350	0.01	0.027
2004-0125		1.25	44,000	600	0.015	0.045	42,000	550	0.015	0.045	40,000	500	0.013	0.036	36,000	350	0.01	0.027
2004-015		1.5	44,000	600	0.015	0.045	42,000	550	0.015	0.045	40,000	500	0.013	0.036	36,000	350	0.01	0.027
2004-020		2	44,000	600	0.015	0.045	42,000	550	0.015	0.045	40,000	500	0.013	0.036	36,000	350	0.01	0.027
2004-025		2.5	35,200	420	0.011	0.034	33,600	385	0.011	0.034	32,000	350	0.01	0.027	28,800	250	0.008	0.02
2004-030		3	35,200	330	0.008	0.024	33,600	310	0.008	0.024	32,000	280	0.008	0.022	28,000	200	0.006	0.016
2004-040		4	35,200	270	0.006	0.019	30,240	250	0.006	0.019	28,800	220	0.006	0.018	25,200	150	0.004	0.012
2005-010	R0.25	1	44,000	900	0.02	0.065	40,000	800	0.015	0.05	36,000	600	0.015	0.05	30,000	400	0.015	0.03
2005-015		1.5	44,000	900	0.02	0.065	40,000	800	0.015	0.05	36,000	600	0.015	0.05	30,000	400	0.015	0.03
2005-020		2	44,000	900	0.02	0.065	40,000	800	0.015	0.05	36,000	600	0.015	0.05	30,000	400	0.015	0.03
2005-025		2.5	44,000	900	0.02	0.065	40,000	800	0.015	0.05	36,000	600	0.015	0.05	30,000	400	0.015	0.03
2005-030		3	32,700	450	0.01	0.04	31,500	400	0.01	0.03	30,000	300	0.008	0.03	24,000	200	0.007	0.015
2005-040		4	32,700	450	0.01	0.04	31,500	400	0.01	0.03	30,000	300	0.008	0.03	24,000	200	0.007	0.015
2005-050		5	26,160	360	0.005	0.02	25,200	320	0.005	0.02	24,000	240	0.004	0.02	19,200	160	0.003	0.01
2005-060		6	26,160	360	0.005	0.02	25,200	320	0.005	0.02	24,000	240	0.004	0.02	19,200	160	0.003	0.01
2006-010	R0.3	1	40,000	1,400	0.045	0.15	36,000	1,500	0.03	0.13	32,000	1,000	0.02	0.1	25,000	600	0.02	0.1
2006-015		1.5	40,000	1,400	0.03	0.13	36,000	1,300	0.03	0.13	32,000	1,000	0.02	0.1	25,000	600	0.02	0.1
2006-020		2	40,000	1,400	0.03	0.13	36,000	1,300	0.03	0.13	32,000	1,000	0.02	0.1	25,000	600	0.02	0.1
2006-025		2.5	40,000	1,200	0.025	0.1	36,000	1,100	0.025	0.1	32,000	900	0.02	0.1	25,000	500	0.02	0.1
2006-030		3	40,000	1,200	0.025	0.1	36,000	1,100	0.025	0.1	32,000	900	0.02	0.1	25,000	500	0.02	0.1
2006-040		4	40,000	1,000	0.02	0.08	32,000	800	0.02	0.08	32,000	700	0.015	0.07	25,000	400	0.01	0.075
2006-050		5	32,000	600	0.01	0.07	28,000	600	0.01	0.05	25,600	500	0.01	0.05	20,000	300	0.005	0.05
2006-060		6	32,000	600	0.01	0.07	28,000	600	0.01	0.05	25,600	500	0.01	0.05	20,000	300	0.005	0.05
2006-080		8	25,600	480	0.008	0.02	22,400	480	0.008	0.02	20,480	350	0.007	0.02	16,000	210	0.004	0.01

HGLB Milling Conditions

WORK MATERIAL			PREHARDENED STEELS / HARDENED STEELS NAK / STAVAX (~55HRC)				HARDENED STEELS SKD11 (55~62HRC)				HARDENED STEELS HAP10 (62~68HRC)				HARDENED STEELS HAP72 (68~70HRC)			
Model Number	Radius of Ball Nose (mm)	Effective Length (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a _p Axial Depth (mm)	a _e Radial Depth (mm)
2008-020	R0.4	2	35,000	1,600	0.06	0.21	30,000	1,600	0.04	0.17	26,000	1,350	0.04	0.15	20,000	700	0.02	0.12
2008-040		4	35,000	1,600	0.06	0.21	30,000	1,600	0.04	0.17	26,000	1,350	0.04	0.15	20,000	700	0.02	0.12
2008-060		6	28,000	1,000	0.02	0.12	25,000	1,000	0.02	0.12	20,800	675	0.02	0.075	16,000	350	0.01	0.06
2008-080		8	22,400	800	0.02	0.1	20,000	800	0.02	0.1	16,640	540	0.016	0.06	12,800	300	0.008	0.048
2010-020	R0.5	2	30,000	1,750	0.2	0.4	24,000	2,000	0.1	0.3	21,000	1,750	0.05	0.2	16,000	875	0.05	0.2
2010-025		2.5	30,000	1,750	0.2	0.4	24,000	2,000	0.1	0.3	21,000	1,750	0.05	0.2	16,000	875	0.05	0.2
2010-030		3	30,000	1,750	0.1	0.3	24,000	2,000	0.1	0.2	21,000	1,750	0.03	0.17	16,000	875	0.03	0.17
2010-040		4	30,000	1,750	0.1	0.3	24,000	2,000	0.1	0.2	21,000	1,750	0.03	0.17	16,000	875	0.03	0.17
2010-050		5	30,000	1,750	0.1	0.3	24,000	2,000	0.1	0.2	21,000	1,750	0.03	0.17	16,000	875	0.03	0.17
2010-060		6	30,000	1,150	0.06	0.23	21,500	1,250	0.03	0.17	19,700	1,050	0.025	0.15	14,500	525	0.025	0.15
2010-080		8	24,000	800	0.025	0.155	18,500	580	0.015	0.12	18,400	480	0.015	0.12	13,800	240	0.015	0.12
2010-100		10	22,000	600	0.018	0.13	14,800	430	0.01	0.09	14,700	360	0.01	0.09	14,700	360	0.01	0.09
2010-120		12	14,150	320	0.015	0.12	13,400	380	0.008	0.08	13,300	290	0.008	0.08	13,300	290	0.008	0.08
2010-140		14	13,500	280	0.012	0.1	12,000	350	0.007	0.08	12,000	220	0.007	0.08	12,000	220	0.007	0.08
2015-030	R0.75	3	30,000	2,450	0.25	0.55	17,000	2,000	0.12	0.4	15,000	1,750	0.06	0.29	11,250	875	0.06	0.29
2015-040		4	30,000	2,450	0.25	0.55	17,000	2,000	0.12	0.4	15,000	1,750	0.06	0.29	11,250	875	0.06	0.29
2015-060		6	30,000	2,450	0.15	0.45	17,000	2,000	0.07	0.31	15,000	1,750	0.04	0.24	11,250	875	0.04	0.24
2015-080		8	23,500	1,300	0.1	0.37	15,000	1,250	0.045	0.25	14,000	1,050	0.03	0.21	10,500	525	0.03	0.21
2015-100		10	23,500	1,300	0.1	0.37	15,000	1,250	0.045	0.25	14,000	1,050	0.03	0.21	10,500	525	0.03	0.21
2015-120		12	13,100	480	0.03	0.21	13,000	580	0.02	0.17	13,000	480	0.02	0.17	9,750	240	0.02	0.17
2015-140		14	11,200	400	0.025	0.19	10,900	490	0.015	0.145	10,900	390	0.015	0.145	8,200	190	0.015	0.145
2020-030	R1	3	28,000	2,900	0.3	0.7	14,000	2,100	0.15	0.5	14,700	2,100	0.15	0.35	12,250	1,800	0.08	0.35
2020-040		4	28,000	2,900	0.3	0.7	14,000	2,100	0.15	0.5	14,700	2,100	0.15	0.35	12,250	1,800	0.08	0.35
2020-060		6	28,000	2,900	0.2	0.6	14,000	2,100	0.1	0.4	14,700	2,100	0.15	0.3	12,250	1,800	0.06	0.3
2020-080		8	28,000	2,900	0.2	0.6	14,000	2,100	0.1	0.4	14,700	2,100	0.15	0.3	12,250	1,800	0.06	0.3
2020-100		10	28,000	2,900	0.2	0.6	14,000	2,100	0.1	0.4	14,700	2,100	0.15	0.3	12,250	1,800	0.06	0.3
2020-120		12	19,500	1,350	0.12	0.45	12,400	1,350	0.06	0.34	13,800	1,320	0.09	0.27	11,500	1,100	0.045	0.27
2020-140		14	19,500	1,350	0.12	0.45	12,400	1,350	0.06	0.34	13,800	1,320	0.09	0.27	11,500	1,100	0.045	0.27
2020-160		16	10,800	500	0.05	0.3	10,800	600	0.03	0.24	12,840	588	0.06	0.24	10,700	490	0.03	0.24
2020-200		20	10,800	500	0.035	0.25	10,800	450	0.02	0.19	10,270	440	0.04	0.19	8,560	370	0.02	0.19
2030-060	R1.5	6	21,000	3,000	0.4	1	13,250	2,740	0.29	0.7	11,040	2,280	0.24	0.55	9,200	1,900	0.12	0.55
2030-080		8	21,000	3,000	0.4	1	13,250	2,740	0.29	0.7	11,040	2,280	0.24	0.55	9,200	1,900	0.12	0.55
2030-100		10	21,000	3,000	0.3	0.9	13,250	2,740	0.24	0.65	11,040	2,280	0.2	0.5	9,200	1,900	0.1	0.5
2030-120		12	21,000	3,000	0.3	0.9	13,250	2,740	0.24	0.65	11,040	2,280	0.2	0.5	9,200	1,900	0.1	0.5
2030-140		14	21,000	3,000	0.3	0.9	13,250	2,740	0.24	0.65	11,040	2,280	0.2	0.5	9,200	1,900	0.1	0.5
2030-160		16	21,000	3,000	0.3	0.9	13,250	2,740	0.24	0.65	11,040	2,280	0.2	0.5	9,200	1,900	0.1	0.5
2030-200		20	14,500	1,600	0.18	0.7	12,380	1,660	0.18	0.5	10,320	1,380	0.15	0.45	8,600	1,150	0.075	0.45
2030-250		25	11,600	1,280	0.14	0.56	11,520	730	0.12	0.38	9,600	610	0.1	0.38	8,000	510	0.05	0.38
2030-300		30	9,280	1,020	0.11	0.45	8,200	530	0.07	0.29	6,840	440	0.06	0.29	5,700	370	0.03	0.29

HGLB Milling Conditions

WORK MATERIAL			PREHARDENED STEELS / HARDENED STEELS NAK / STAVAX (~55HRC)				HARDENED STEELS SKD11 (55~62HRC)				HARDENED STEELS HAP10 (62~68HRC)				HARDENED STEELS HAP72 (68~70HRC)			
Model Number	Radius of Ball Nose (mm)	Effective Length (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)
2040-080	R2	8	18,000	3,200	0.5	1.3	11,380	2,880	0.36	0.95	9,480	2,400	0.3	0.75	7,900	2,000	0.15	0.75
2040-100		10	18,000	3,200	0.5	1.3	11,380	2,880	0.36	0.95	9,480	2,400	0.3	0.75	7,900	2,000	0.15	0.75
2040-120		12	18,000	3,200	0.4	1.2	11,380	2,880	0.31	0.85	9,480	2,400	0.26	0.7	7,900	2,000	0.13	0.7
2040-140		14	18,000	3,200	0.4	1.2	11,380	2,880	0.31	0.85	9,480	2,400	0.26	0.7	7,900	2,000	0.13	0.7
2040-160		16	18,000	3,200	0.4	1.2	11,380	2,880	0.31	0.85	9,480	2,400	0.26	0.7	7,900	2,000	0.13	0.7
2040-200		20	18,000	3,200	0.4	1.2	10,730	1,800	0.21	0.7	8,940	1,500	0.18	0.55	7,450	1,250	0.09	0.55
2040-250		25	12,500	1,250	0.25	0.95	10,730	1,800	0.21	0.7	8,940	1,500	0.18	0.55	7,450	1,250	0.09	0.55
2040-300		30	10,630	1,000	0.2	0.76	10,080	780	0.15	0.45	8,400	650	0.12	0.45	7,000	540	0.06	0.45
2040-350		35	9,030	800	0.16	0.61	8,640	730	0.13	0.43	7,200	610	0.11	0.43	6,000	510	0.055	0.43
2060-100	R3	10	14,400	3,200	0.5	1.5	9,140	2,880	0.38	1.05	7,620	2,400	0.32	0.88	6,350	2,000	0.16	0.88
2060-150		15	14,400	3,200	0.5	1.5	9,140	2,880	0.38	1.05	7,620	2,400	0.32	0.88	6,350	2,000	0.16	0.88
2060-200		20	14,400	3,200	0.5	1.5	9,000	2,300	0.32	0.95	7,620	2,400	0.32	0.88	6,350	2,000	0.16	0.88
2060-250		25	14,400	3,200	0.5	1.5	8,100	2,000	0.3	0.95	7,500	1,920	0.27	0.805	6,250	1,600	0.135	0.805
2060-300		30	14,400	3,200	0.5	1.5	7,700	1,800	0.26	0.88	7,440	1,500	0.22	0.73	6,200	1,250	0.11	0.73



a_p : Axial Depth (mm)
 a_e : Radial Depth (mm)

Note:

- Decrease the feed rate more than 50% from the milling parameters when slot milling.
- Decrease both spindle speed and feed rate proportionally when the milling parameters exceed the machine's maximum spindle speed, or when chattering and red-hot occur.
- Every coolant offers stable milling.

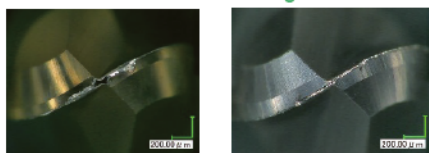
Wear Comparison

HGLB R1 × Effective Length 6

HAP72 (69HRC)

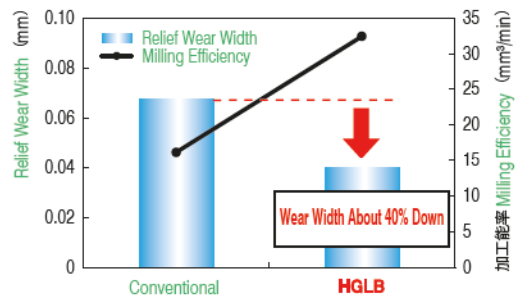
HGLB mills twice as efficient as the conventional tool.

Tools after milling



Conventional

HGLB



Tool	Conventional	HGLB 2020-060
Spindle Speed	9,200 min ⁻¹	12,250 min ⁻¹
Feed Rate	900 mm/min	1,800 mm/min
Axial Depth a_p	0.06 mm	
Radial Depth a_e	0.3 mm	
Coolant	Air Blow (Through-Spindle)	
Milling Shape	Square Pocket (20 x 15 x Depth 2 mm) × 2 Pockets	
Cycle Time	76 min	50 min

$$\text{Milling Efficiency (mm}^3\text{/min)} = \text{Feed Rate} \times a_p \times a_e$$



Advisory for Safe Use of UNIMAX Tungsten Carbide End Mills

Correct application and operation is strongly advised to avoid clogging, abrasion, etc., that could cause serious accidents or injuries. Ignition or sparks generated during milling could lead to fire or extreme damage to the work piece. End Mills are made with very sharp cutting edges and must be handled with extra care.

- * Never touch the cutting edge with your bare hands, as this could cause serious injury. Special caution is required when opening the package.
- * Dropping the tool could cause breakage or flying debris, leading to serious injury.
- * During milling, unexpected impact or shock on the tool could cause breakage or flying debris. Ensure to use protective items such as safety glasses and a face guard.
- * For best results, fine parameter adjustment may be required, depending on the materials; milling shape and strategy; machine rigidity and spindle capability.
- * Use a machine that has high rigidity and generates a low level of vibration.
- * Do not use flammable cutting oils.

Advisory for regrinding UNIMAX Tungsten Carbide End Mills

- * Never grind the tool without wearing safety glasses and a face guard.



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Price & Specifications are subject to change without notice.

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